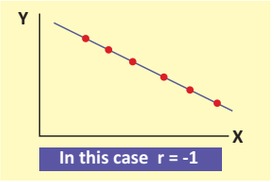
5) Perfect Negative correlation

If all the plotted points lie on a straight line falling from upper left hand corner to lower right hand corner, the two variables have perfect negative correlation.



4.3.1 Merits and Demerits of scatter diagram

Merits

- It is a simple and non-mathematical method of studying correlation between the variables.
- It is not influenced by the extreme items
- It is the first step in investigating the relationship between two variables.
- It gives a rough idea at a glance whether there is a positive correlation, negative correlation or uncorrelated.

Demerits

- We get an idea about the direction of correlation but we cannot establish the exact strength of correlation between the variables.
- No mathematical formula is involved.

4.4 KARL PEARSON'S CORRELATION COEFFICIENT

When there exists some relationship between two measurable variables, we compute the degree of relationship using the correlation coefficient.

Co-variance

Let (X,Y) be a bivariable normal random variable where $V(X)$ and $V(Y)$ exists. Then, covariance between X and Y is defined as

$$\text{cov}(X,Y) = E[(X-E(X))(Y-E(Y))] = E(XY) - E(X)E(Y)$$

If (x_i,y_i), i=1,2, ..., n is a set of n realisations of (X,Y), then the sample covariance between X and Y can be calculated from

$$\text{cov}\left(X,Y\right)=\frac{1}{n}\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y})=\frac{1}{n}\sum_{i=1}^{n}x_iy_i-\bar{x}\,\bar{y}$$

4.4.1 Karl Pearson's coefficient of correlation

When X and Y are linearly related and (X,Y) has a bivariate normal distribution, the co-efficient of correlation between X and Y is defined as

$$r\left(X,Y\right)=\frac{\text{cov}(X,Y)}{\sqrt{V(X)V(Y)}}$$

This is also called as product moment correlation co-efficient which was defined by Karl Pearson.

Based on a given set of n paired observations (x_i,y_i), i=1,2, ... n the sample correlation co-efficient between X and Y can be calculated from

$$r\left(X,Y\right)=\frac{\frac{1}{n}\sum_{i=1}^{n}x_iy_i-\bar{x}\;\bar{y}}{\sqrt{\frac{1}{n}\sum_{i=1}^{n}x_i^2-\bar{x}^2}\sqrt{\frac{1}{n}\sum_{i=1}^{n}y_i^2-\bar{y}^2}}$$

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